WES 265 HW 1 - Problems 2 & 3

Problem 2 - Linear Transformations of Random Variables

$$Z = aX + b$$

1. $$E[Z] = E[aX + b]$$

$$= aE[X] + b$$

$$= a\mu + b$$

2. $$Var(Z) = Var(aX + b)$$

$$= a^2 Var(X)$$

$$= a^2\sigma^2$$

3. $$Var(Z) = E[Z^2] - (E[Z])^2$$

$$= E[(aX+b)^2] - (a\mu + b)^2$$

$$= a^2 E[X^2] + b^2 + 2aE[X]b$$

$$- (a\mu + b)^2$$

$$= a^2 E[x^2] + \cancel{b^2} + \cancel{2a\mu b} - a\mu$$

$$- \cancel{b^2} - \cancel{2a\mu b}$$

$$= a^2 \left(E[x^2] - \mu^2 \right)$$

$$= a^2 \sigma^2$$

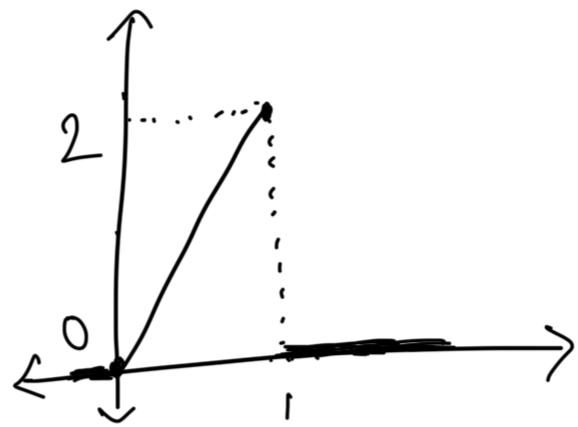
Problem 3 - Expectation and Variance from a PDF

$$P_X(x) = \begin{cases} 2x & x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

1. a) PDF must be non-negative

$$P_X(x) \geq 0 \quad \forall \; x \in (-\infty, \infty)$$

$P_X(x)$:



This condition is clearly satisfied.

b) PDF must integrate to 1.

$$\int_{-\infty}^{\infty} P_X(x)\,dx = 1$$

$$\int_{0}^{1} (2x)\,dx = \left[x^2\right]_0^1 = 1$$

2. $$E[X] = \int_{-\infty}^{\infty} x \, p_X(x) \, dx$$

$$\Rightarrow E[X] = \int_0^1 x \cdot 2x \cdot dx = 2 \int_0^1 x^2 \cdot dx$$

$$= 2 \left[\frac{x^3}{3} \right]_0^1 = \frac{2}{3}$$

3. $$Var(X) = E[X^2] - (E[X])^2$$

$$\Rightarrow Var(X) = \int_{-\infty}^{\infty} x^2 \cdot p_X(x) \cdot dx - \left(\frac{2}{3}\right)^2$$

$$= \int_0^1 x^2 \cdot 2x \cdot dx - \left(\frac{2}{3}\right)^2$$

$$= 2 \int_0^1 x^3 \cdot dx - \frac{4}{9}$$

$$= 2\left[\frac{x^4}{4}\right]_0^1 - \frac{4}{9} = \frac{2}{4} - \frac{4}{9}$$

$$= \frac{1}{18}$$